



1.3 Opportunity Cost and Production Possibilities Curve

* Learning Objectives:

1. Calculate the opportunity cost given a decision made
2. Explain what is Production Possibilities Curve and how it will shift
3. Draw Production Possibilities Curve

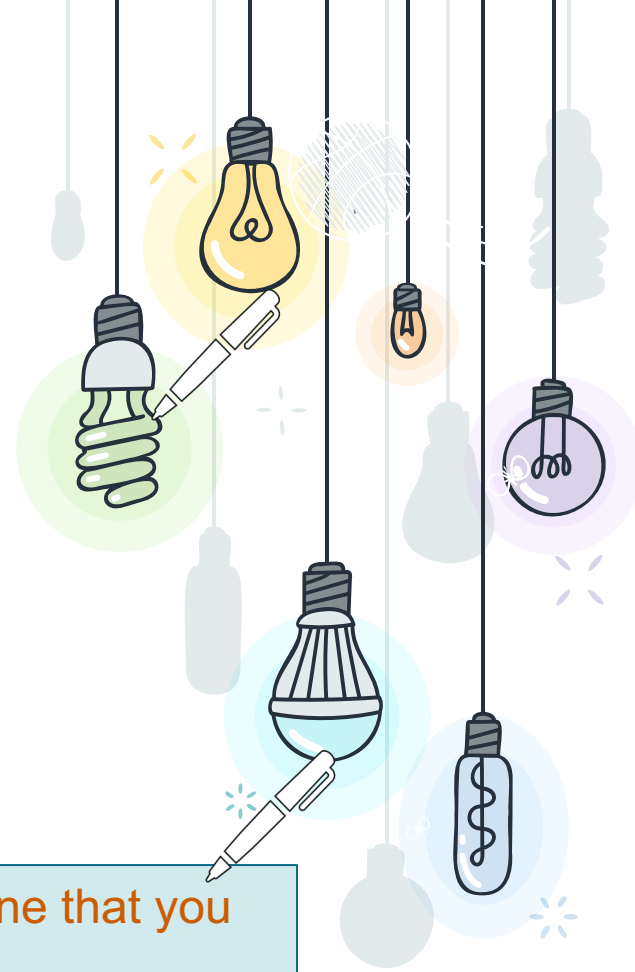


* Definition:

Opportunity cost is the best alternative that we give up, when we make a choice or a decision.

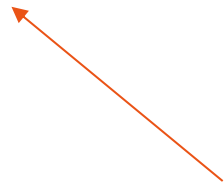
Words:

Alternative: An alternative plan or method is one that you can use if you do not want to use another one.

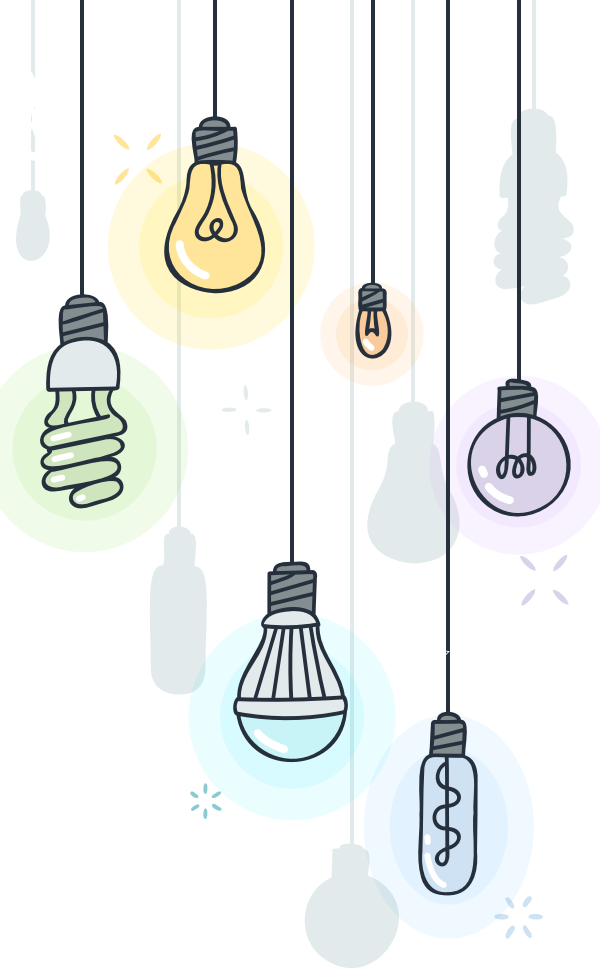




Calculate



HOW TO MEASURE Opportunity COST



JACK'S OPPORTUNITY COST

cars	bicycles
4	6
$\frac{4}{4} = 1$	$\frac{6}{4} = 1.5$

What is his opportunity cost of producing 4 cars?

His opportunity cost of producing 4 cars is 6 bicycles.

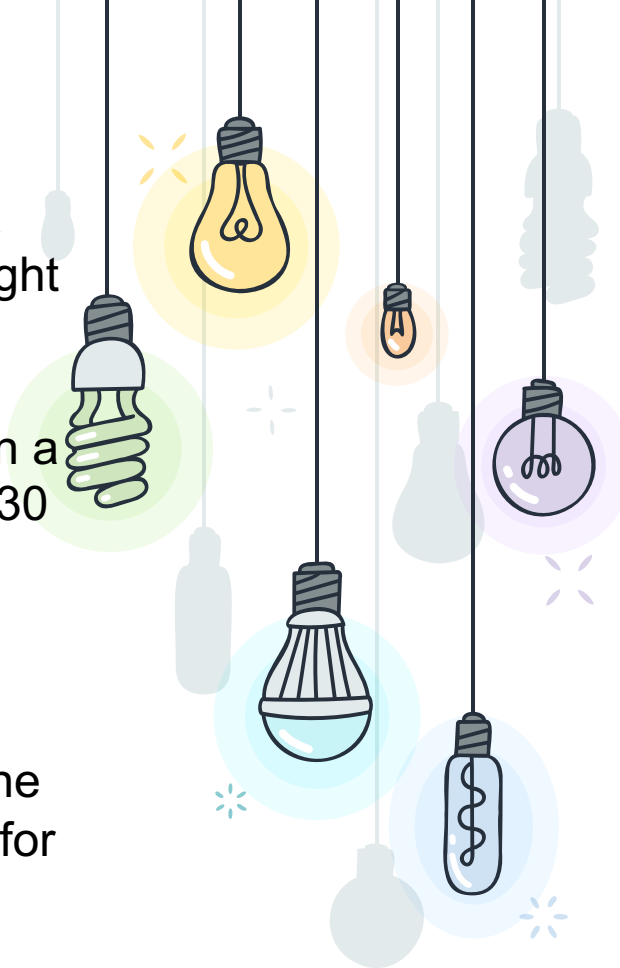
What is his opportunity cost of producing one car?

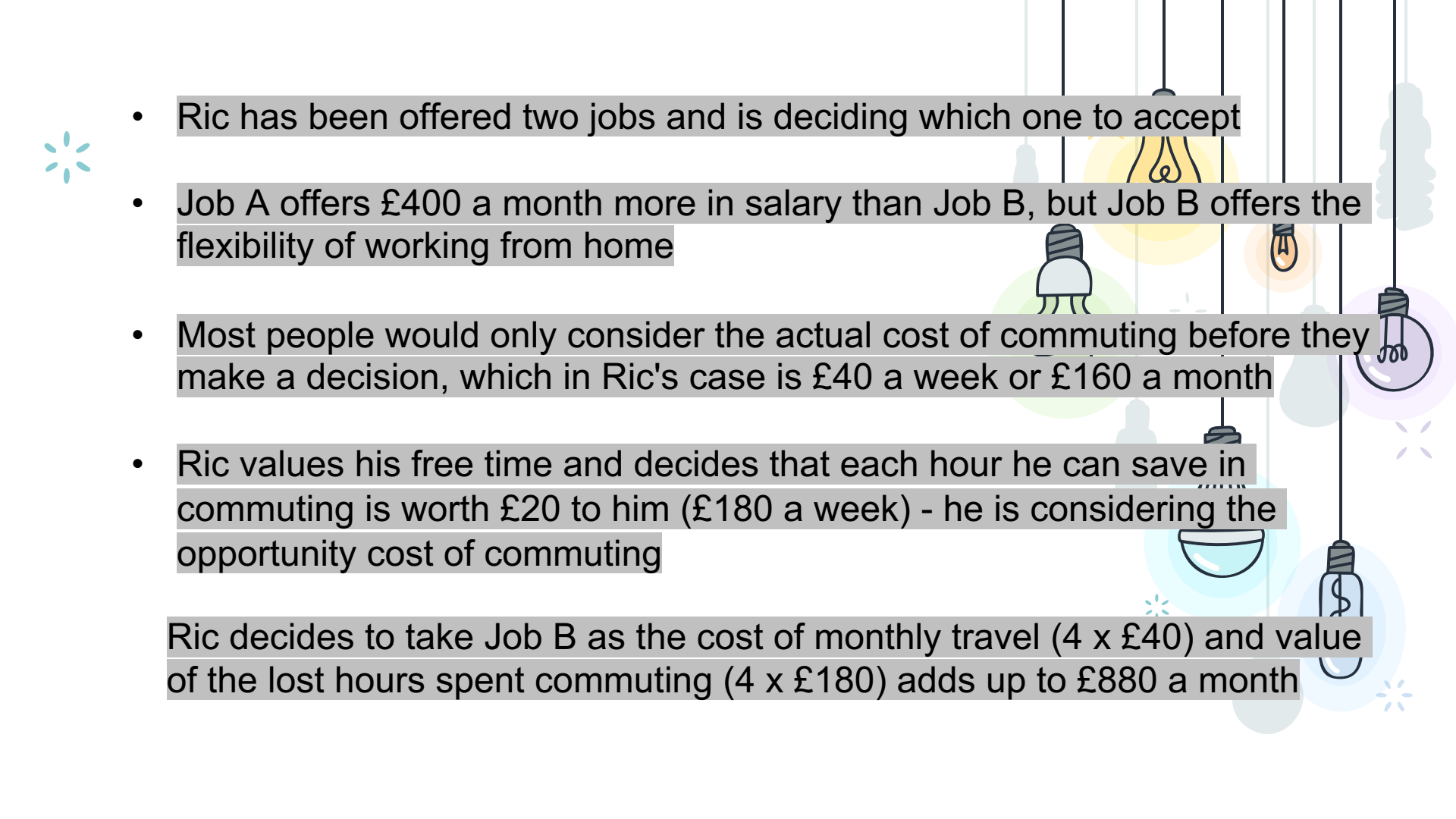
His opportunity cost of producing 1 car is 1.5 bicycles.



- Ashika is wanting to visit her best friend in Iceland
- She looks at flight prices from London to Reykjavík
- On Friday night it costs £120 whereas Thursday night is only £50
- She is about to book the Thursday flight but then realises that the **opportunity cost** of saving £60 on a flight is the **inability to work on Friday** (loss of £130 income)

Ashika books the more expensive flight. If she had booked the cheaper flight, it would have cost her the income from the missed day of work (£130) + £50 for the ticket



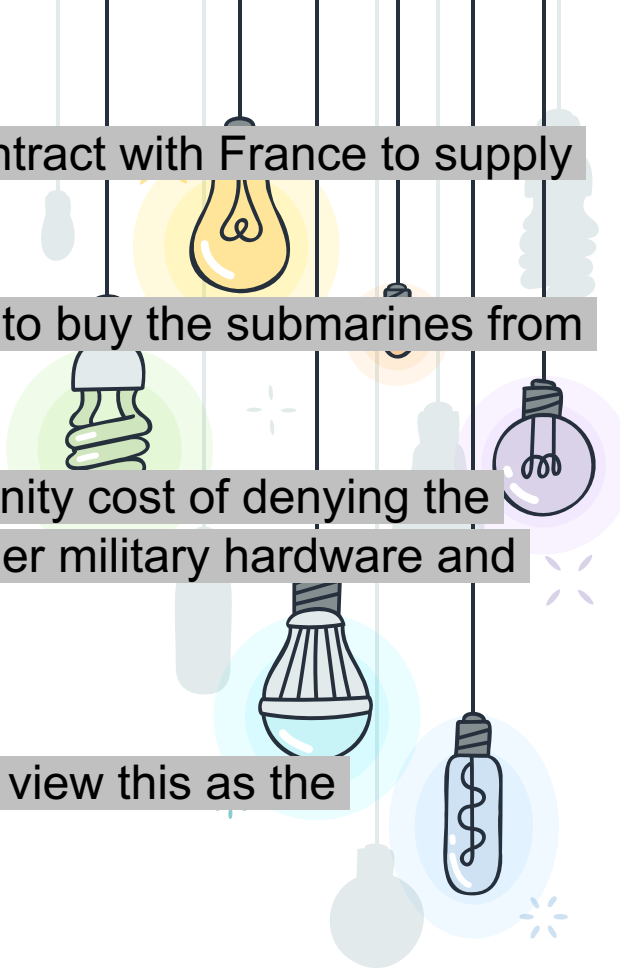
- 
- Ric has been offered two jobs and is deciding which one to accept
 - Job A offers £400 a month more in salary than Job B, but Job B offers the flexibility of working from home
 - Most people would only consider the actual cost of commuting before they make a decision, which in Ric's case is £40 a week or £160 a month
 - Ric values his free time and decides that each hour he can save in commuting is worth £20 to him (£180 a week) - he is considering the opportunity cost of commuting

Ric decides to take Job B as the cost of monthly travel ($4 \times £40$) and value of the lost hours spent commuting ($4 \times £180$) adds up to £880 a month



- The Australian Government has entered into a contract with France to supply them with 8 submarines valued at \$70 billion
- The USA hears about it and pressurises Australia to buy the submarines from them instead
- The Australian Government considers the opportunity cost of denying the USA, which includes less preferential deals on other military hardware and general trade agreements

They decide to break the contract with France and view this as the approach that carries the lowest opportunity cost



The cost of making choices

A factory employs 10 people and has 2 machines able to produce and pack 300 glass bottles each day. The same employees and machines could instead be used in the same factory to make and pack 400 glass jars each day.

The same 10 employees used to work on a farm and were very skilled at growing and harvesting corn until the farm was sold and the factory was built on the farmland. Compared to the farm the factory is noisy and pollutes the air with smoke.

- What is the opportunity cost to the factory owners of using the 10 employees and 2 machines to make 300 bottles each day?

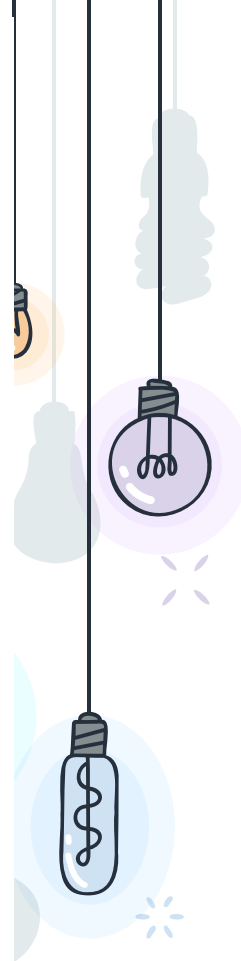
The profit that making 400 glass jars can bring.

- What is the opportunity cost to the employees of working in the factory?

The income that the employees can get by working on the farm.

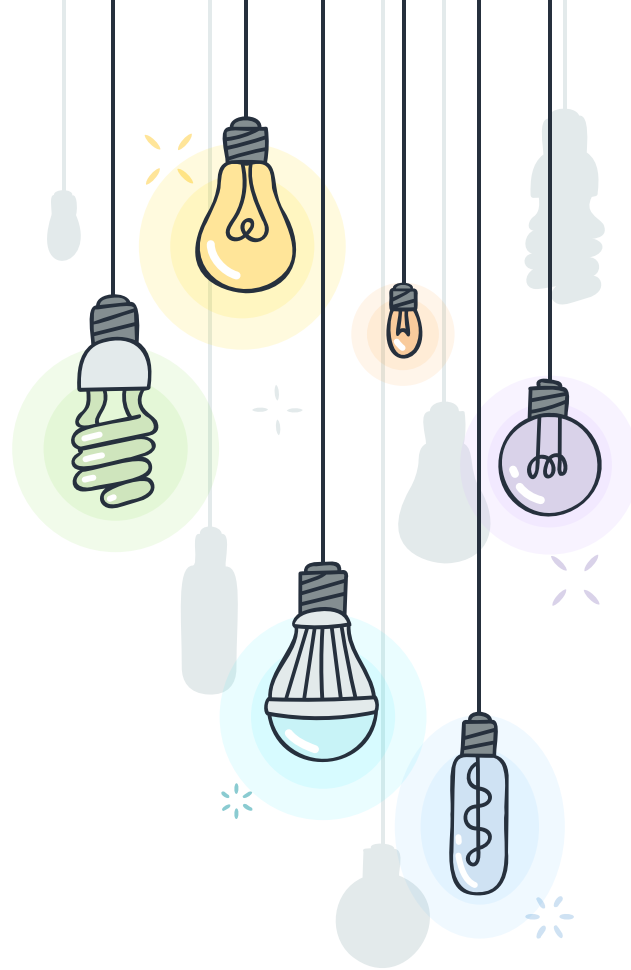
- What is the opportunity cost to society of the factory?

The benefits that farms can bring to society, such as a good environment, less noise and clean air.

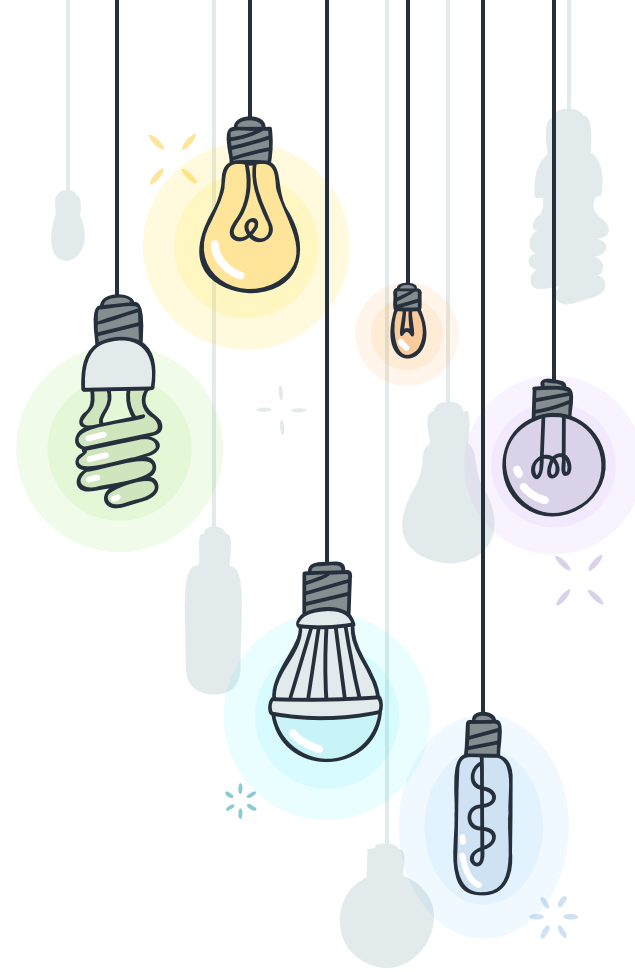




Why WE NEED TO LEARN Opportunity COST



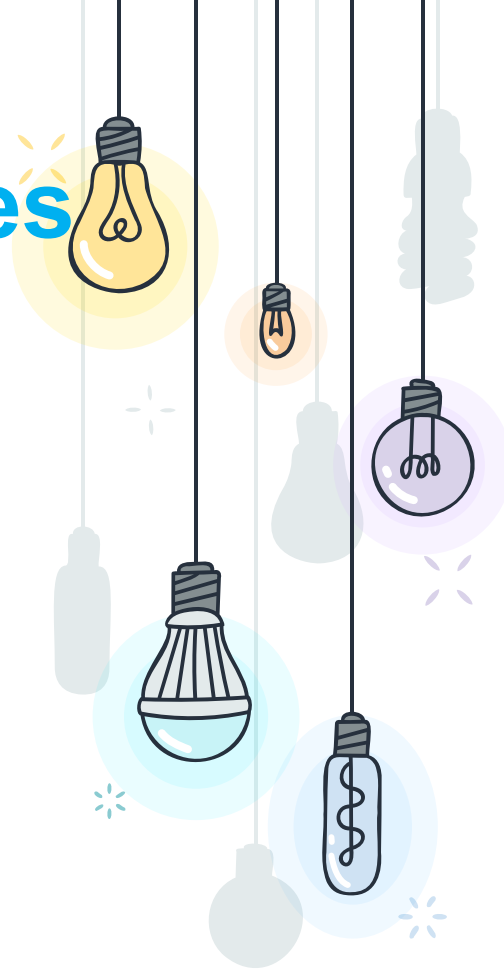
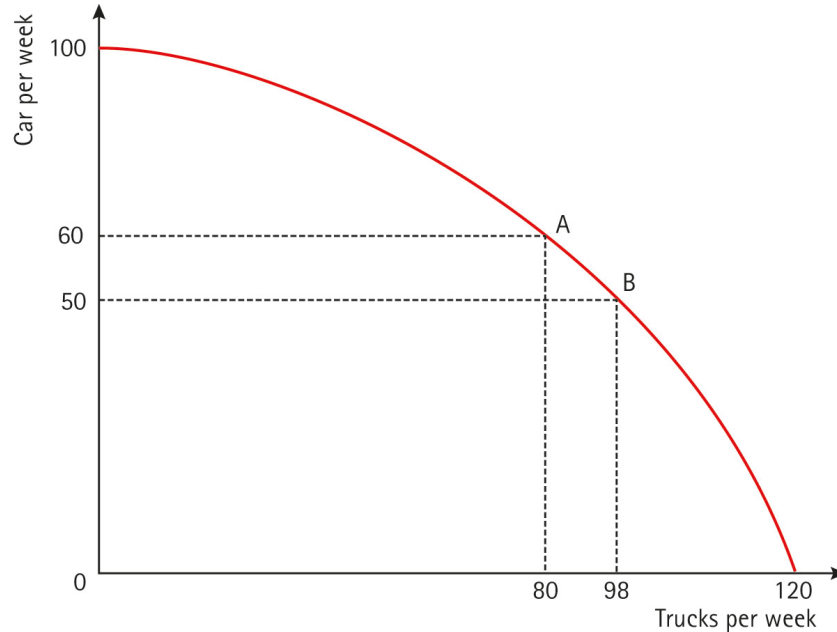
PRODUCTION **p**OSSIBILITY CURVES



* Production possibility curves

A firm producing cars and trucks

What is the opportunity cost of producing 18 more trucks?

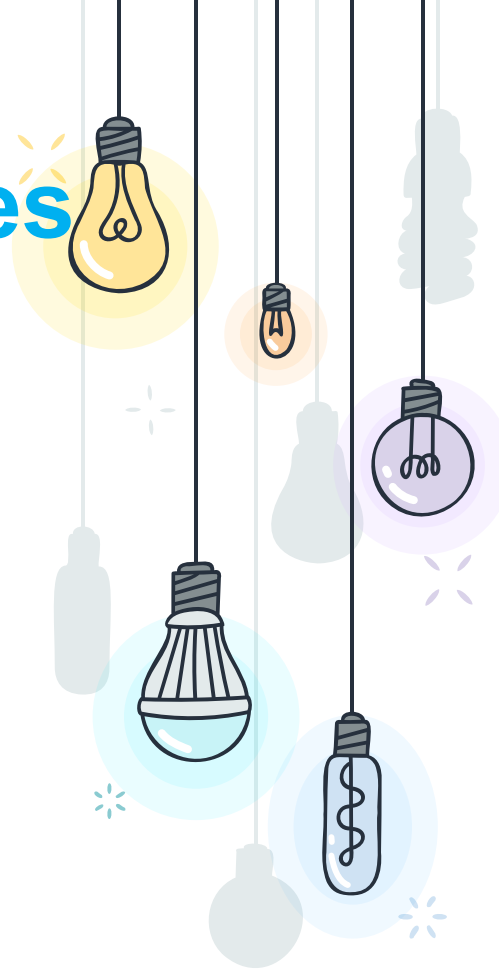


* Production possibility curves

Production possibility curves (PPCs) show the different combinations of two goods being produced using all resources efficiently.

But, because resources are limited, producing more of one product means producing less of another.

PPCs are therefore a useful way of showing the opportunity cost of producing more of one product in terms of how much of another must be given up.

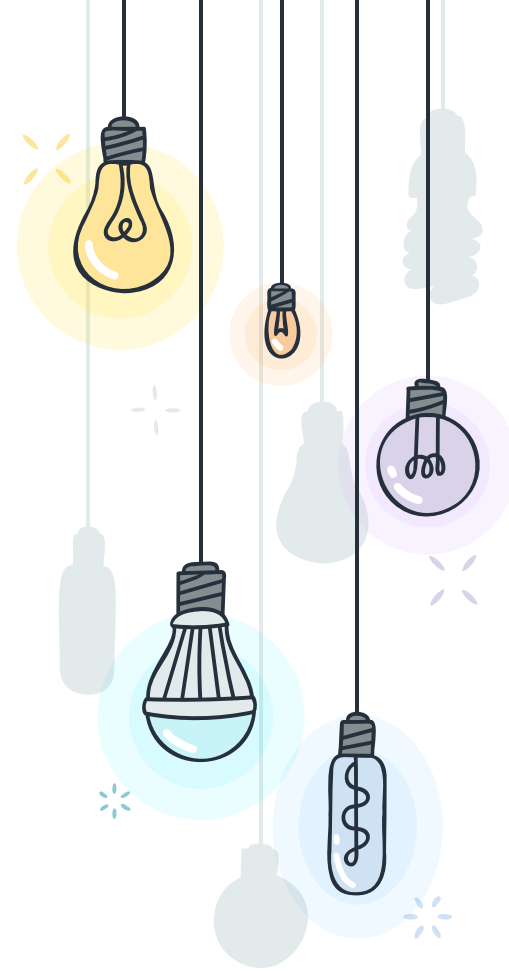




SUBJECT VOCABULARY

capital goods those purchased by firms and used to produce other goods such as factories machinery, tools and equipment

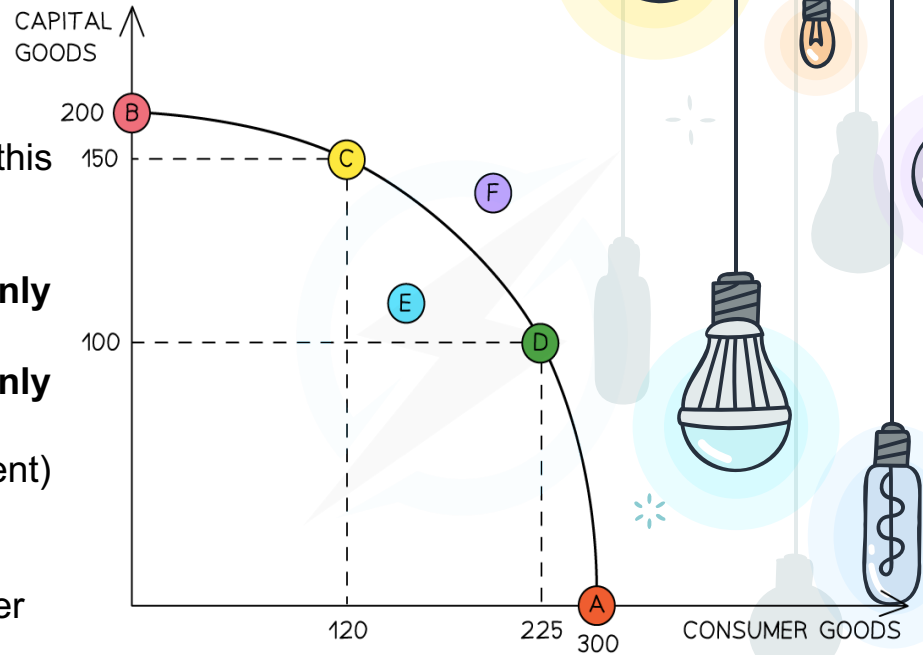
consumer goods those purchased by households such as food, confectionery, cars, tablets and furniture



Production possibility curves

The use of PPC to depict the maximum productive potential of an economy

- The curve demonstrates the **possible combinations of the maximum output** this economy can produce **using all of its resources** (factors of production)
- At A, its resources are used to produce **only consumer goods** (300)
- At B, its resources are used to produce **only capital goods** (200)
- Points C and D both represent **full** (efficient) use of an economy's resources, as these points **fall on the curve**
- At C, 150 capital goods and 120 consumer goods are produced

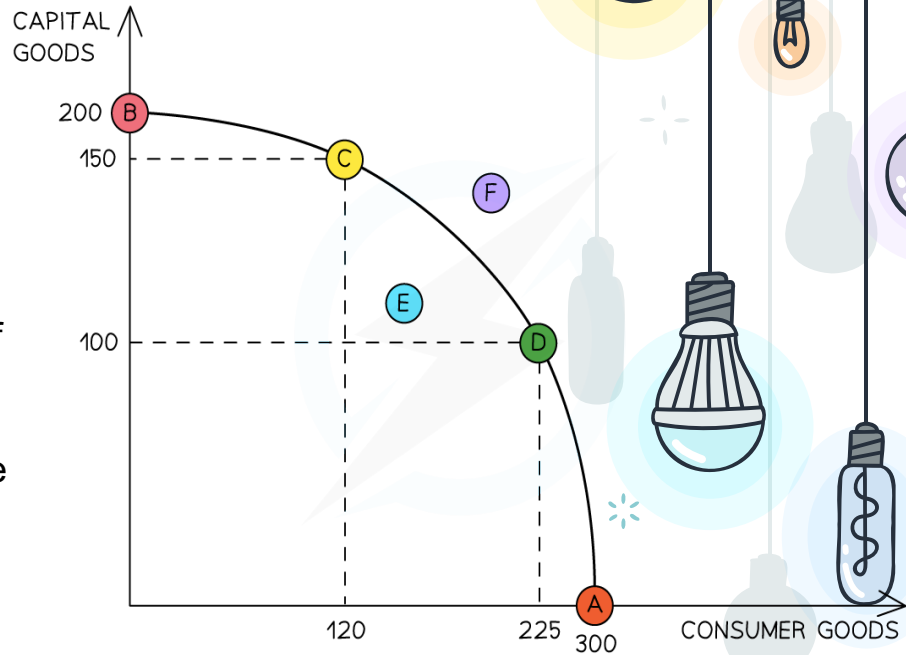


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Production possibility curves

The use of PPC to depict opportunity cost

- To produce one more unit of capital goods, this economy must give up production of some units of consumer goods (limited resources)
- If this economy moves from point C (120, 150) to point D (225, 100), the opportunity cost of producing an additional 105 units of consumer goods is 50 capital goods
- A movement in the PPC occurs when there is any change in the allocation of existing resources within an economy, such as the movement from point C to D



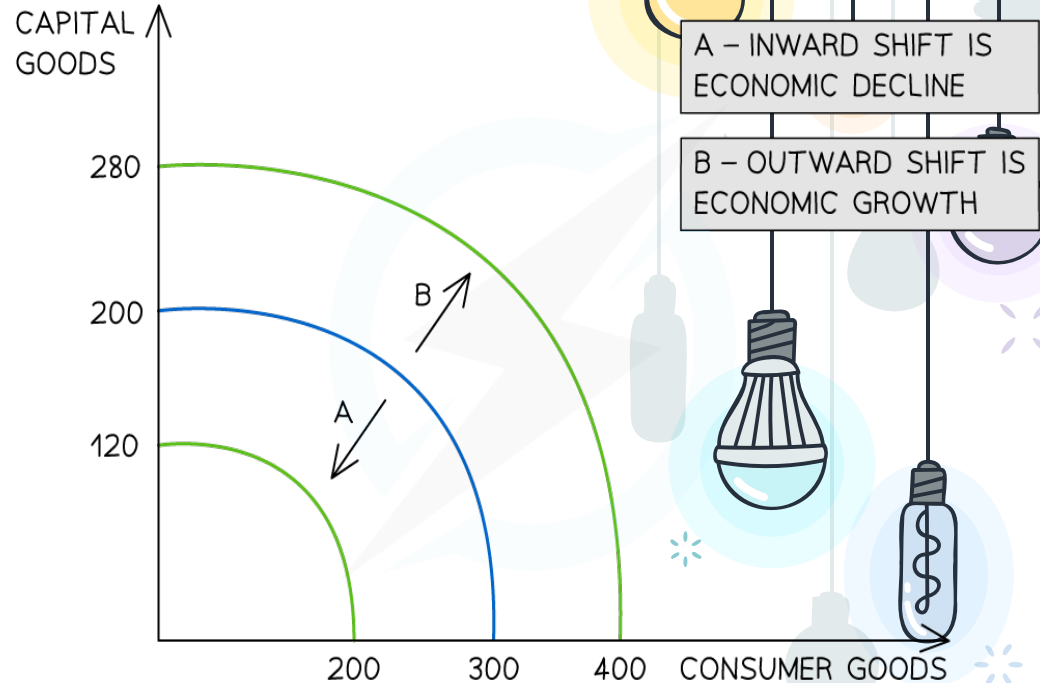
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SHIFTS IN A PPC



The use of PPC to depict opportunity cost

- **Economic growth** occurs when there is an increase in the productive potential of an economy
- This is demonstrated by an outward shift of the entire curve. More consumer goods and more capital goods can now be produced using all of the available resources

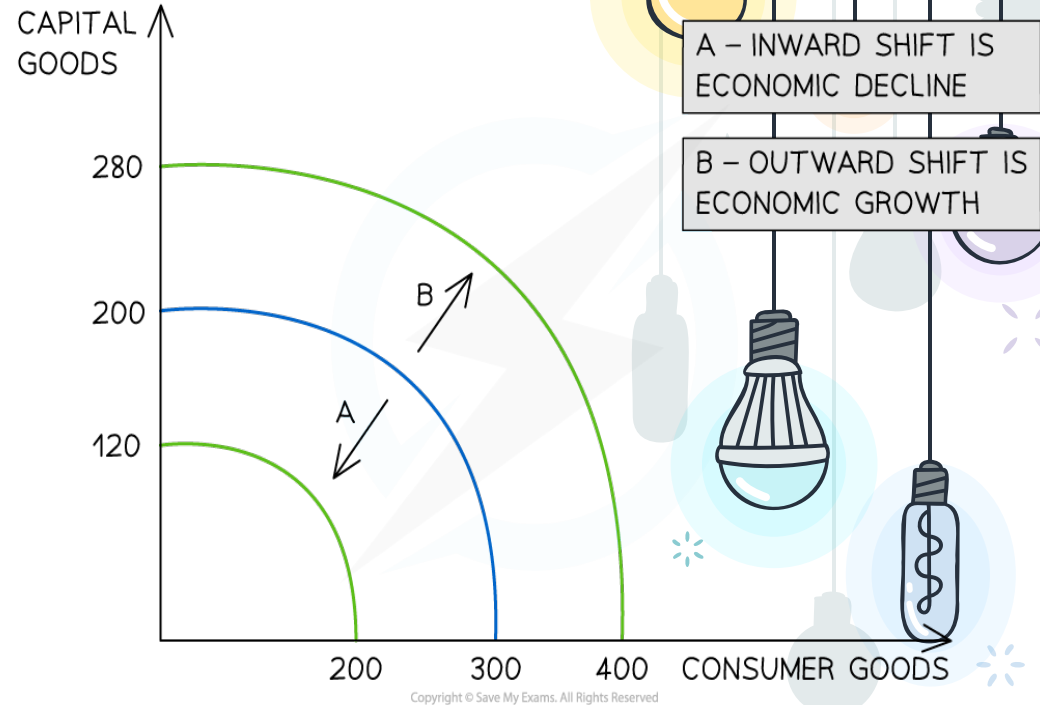


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SHIFTS IN A PPC

The use of PPC to depict opportunity cost

- This shift is caused by an increase in the **quality or quantity** of the available factors of production
- One example of how **the quality** of a factor of production can be improved is through the impact of training and education on labour. An educated workforce is a more productive workforce and the production possibilities increase
- One example of how **the quantity** of a factor of production can be increased is through a change in migration policies. If an economy allows more foreign workers to work productively in the economy, then production possibilities increase

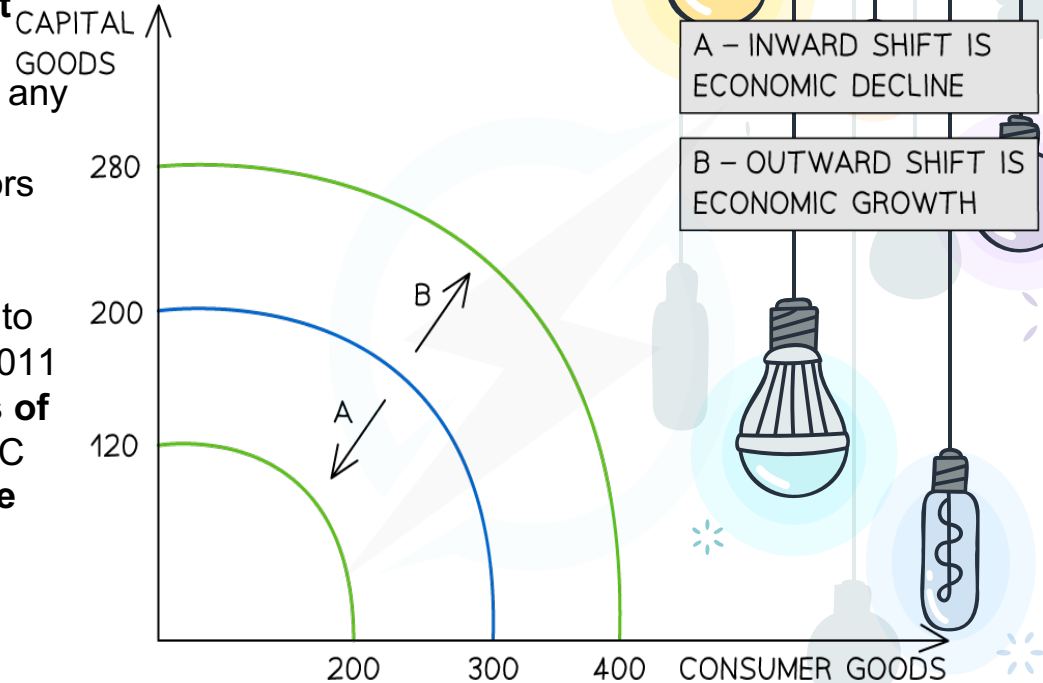


SHIFTS IN A PPC



The use of PPC to depict opportunity cost

- **Economic decline** occurs when there is any impact on an economy that **reduces the quantity or quality** of the available factors of production
- One example of how this may happen is to consider how the Japanese tsunami of 2011 devastated the **production possibilities of Japan** for many years. It shifted their PPC inward and resulted in **economic decline**



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